DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

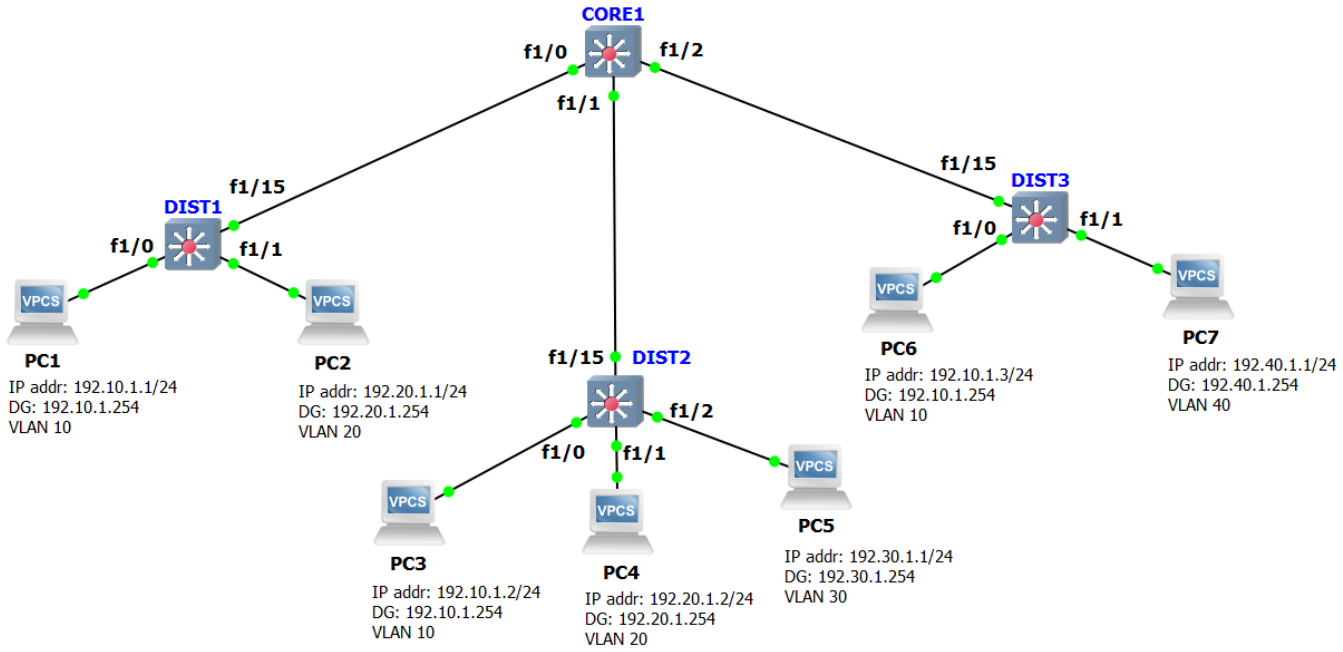
ANO 2025/2026

REDES DE COMUNICAÇÕES II

STUDENTS AUTO-EVALUATION OF LABORATORY GUIDE NO. 2

Scenario 1

Consider the following network with a distributed IP routing approach (i.e., CORE1 is the default gateway of end-to-end VLANs; DIST2 and DIST3 are the default gateways of the corresponding local VLANs) with trunk links supporting the minimum required VLANs. There is an interconnection VLAN and RIPv2 is appropriately configured in CORE1, DIST2 and DIST3 so that there is full IP connectivity.



In this scenario, the routing table of DIST2 is:

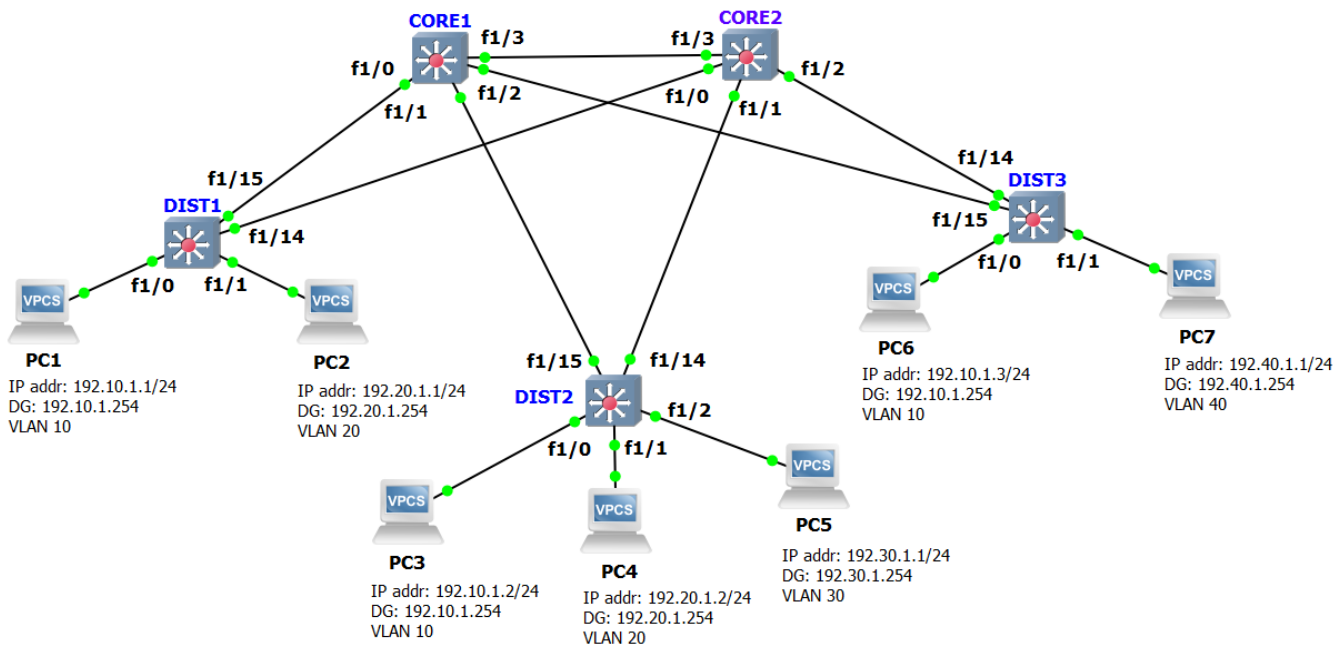
```
R    192.20.1.0/24 [120/1] via 192.10.10.1, Vlan100
C    192.30.1.0/24 is directly connected, Vlan30
C    192.10.10.0/24 is directly connected, Vlan100
R    192.10.1.0/24 [120/1] via 192.10.10.1, Vlan100
R    192.40.1.0/24 [120/1] via 192.10.10.5, Vlan100
```

Classify as True (T) or False (F) each of the following statements:

- ☐ a) The interconnection VLAN has the IP address 192.10.10.0/25.
- ☐ b) The address 192.20.1.254 is configured in DIST2.
- ☐ c) The address 192.10.10.1 is configured in CORE1.
- ☐ d) In a ping from PC1 to the IP address 192.40.1.1, an ICMP Echo Reply packet in VLAN 10 is captured in link CORE1-DIST3.
- ☐ e) In a ping from PC2 to the IP address 192.10.1.1, there are no ICMP packets captured in link DIST1-CORE3.
- ☐ f) In a ping from PC5 to the IP address 192.20.1.3, ARP Request packets are captured in link CORE1-DIST3 in VLAN 20.
- ☐ g) In a ping from PC5 to the IP address 192.10.10.5, an ICMP Echo Reply packet in VLAN 30 is captured in link CORE1-DIST3.
- ☐ h) Link CORE1-DIST1 belongs to the broadcast domain of VLAN 100.

Scenario 2

Consider the following network where both CORE1 and CORE2 are routing the IP networks between all VLANs and where, through VRRP, CORE1 is the default gateway of VLANs 30 and 40 and CORE2 is the default gateway of VLANs 10 and 20. All trunk links support all existing VLANs.



In this scenario, the spanning tree brief information of VLAN 10 in CORE2 is:

```
VLAN10
  Spanning tree enabled protocol ieee
  Root ID    Priority    16384
             Address     c205.43bc.0001
             Cost        19
             Port        44 (FastEthernet1/3)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    32768
             Address     c204.3c28.0001
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300
Interface                                     Designated
Name               Port ID Prio Cost  Sts Cost  Bridge ID            Port ID
----------------- ------- ---- ----- --- ----- -------------------- -------
FastEthernet1/0     128.41  128    19  BLK    10 32768 c202.19b4.0000 128.55
FastEthernet1/1     128.42  128    19  BLK    15 32768 c203.32e0.0000 128.55
FastEthernet1/2     128.43  128    19  BLK    10 32768 c201.196c.0000 128.55
FastEthernet1/3     128.44  128    19  FWD     0 16384 c205.43bc.0001 128.44
```

Classify as True (T) or False (F) each of the following statements concerning VLAN 10:

- [ ] a) The root bridge has priority 16384.
- [ ] b) The port cost of `f1/15` of DIST1 is 19.
- [ ] c) The interface `f1/14` of DIST2 is a root interface.
- [ ] d) One cannot know the port cost of the interface `f1/3` of CORE1.
- [ ] e) The interface `f1/15` of DIST3 is a root interface.

- [ ] f) The bridge ID of DIST2 is 32768 c201.196c.0000.
- [ ] g) Changing the port cost of `f1/2` of CORE2 to 12 does not change the VLAN 10 spanning tree.
- [ ] h) In a ping from PC7 to the IP address 192.10.1.1, a ICMP Echo Request in VLAN 10 is captured in link CORE1-CORE2.
- [ ] i) In a ping from PC3 to the IP address 192.30.1.1, a ICMP Echo Request in VLAN 10 is captured in link CORE1-CORE2.
- [ ] j) In a ping from PC6 to PC1, the ICMP Echo Requests are sent through the path DIST3 → CORE2 → DIST1.